

Summary of the data quality issues

1. Distance log file
 1. Starting point and destination do not follow a standard format. While some records follow the format “city, province”, many records have inconsistent values like “nisku to grande praire”
 2. Destination value mismatch because of the values “nisku to grande praire”
 3. Some places have abbreviations used for starting point and ending point like “Nisku to GP, AB”
 4. No vin number/truck number column
 5. Origin and Destination do not have the corresponding province mentioned. Province level mapping issues
 6. Inconsistent city names like “Edmanton” instead of “Edmonton
 7. Inconsistent date formats
 8. Missing dates
 9. Jurisdiction level distance data missing
 10. Fuel consumption values does not match distance, eg: Caledon - Red Deer (3250 KM) but fuel invoice shows 46.445L
 11. Distance values not consistent to actual values, eg: Burnaby - Langley 85 KM but actual is less than half of it.
2. PDF Fuel invoice file
 1. Every invoice receipts follows a different structure of key value fields. Example: some have price/litre included, some don't.
 2. Tax split for gst/phst/fhst not available as a standard in all invoices.
 3. Tax_payer_ID not available as a standard column in all invoices.
 4. Time value is not extracted by the API and are inconsistent formats.
 5. Litres of fuel charged is missing and consistent across invoices
 6. Fuel type is missing and consistent across invoices
 7. Form of payment is missing and not consistent across invoices
 8. City to province mapping issues not standardized.
 9. Incorrect city names mapped by the API.
 10. Province missing from API extraction.

Assumptions:

1. Confidence levels have not been propagated to the warehouse assuming it is for reference purposes that is available with the Textract output file.
2. As date is an important ID parameter for this use case a default date has been set wherever missing.
3. Data dictionary for city-province mapping limited to sample data usage.